Newcastle University
SCHOOL OF COMPUTING

Bridging the Goal-to-Action Gap

Planning-Time Grounding vs Execution-Time Recovery in LLM Embodied Planning

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github.com/zihanwork/thesis-code

Motivation

Large language models can understand a household goal yet still fail to produce a sequence that is **syntactically valid, executable, and goal-completing**.

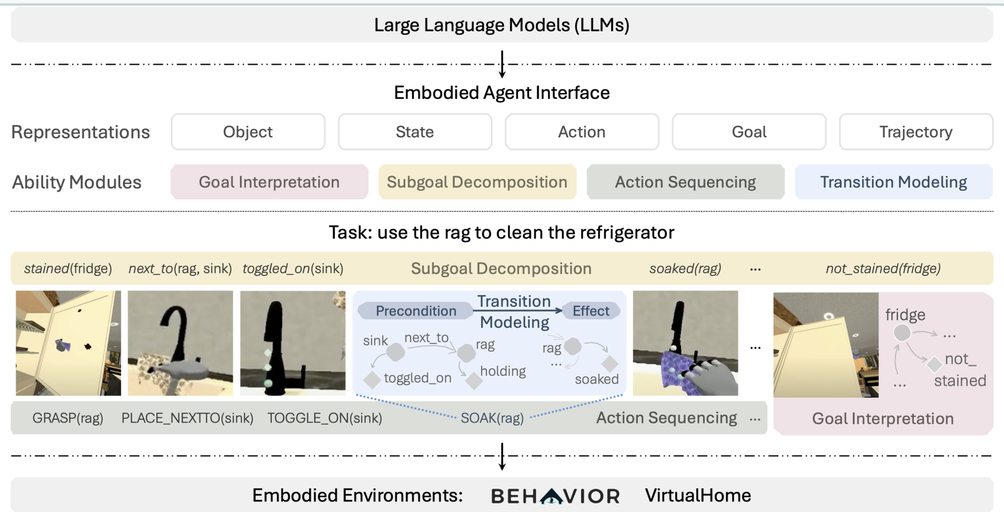
Prior systems often bundle richer prompts, retrieval, validation and repair. When performance improves, it is unclear **where the gain came from**.

> **Under matched tasks and models, how much comes from better initial planning, how much from recovery, and how do the two interact?**

Research Questions

- **RQ1 — Grounding:** Do structured constraints, Flat RAG or GraphRAG improve initial plans?
- **RQ2 — Failure:** Which trajectory and goal failures remain?
- **RQ3 — Recovery & interaction:** How much does each recovery method add within each planner?
- **RQ4 — Transfer:** Are effects consistent across three model families?
- **RQ5 — Cost:** What calls, tokens, latency and search are required?

Benchmark & Scope



Embodied Agent Interface (EAI) links LLMs to symbolic representations and embodied environments. This study isolates VirtualHome Action Sequencing.

84 official-compatible tasks	**8** task families
60 current analysis cells	**7,056** official records generated

Evidence package: 5,040 archived full-grid records plus a 2,016-record P1/P2 replacement replication. The same 84 pre-screened tasks and denominator are used throughout.

Contribution

- A controlled **two-axis factorial study** separating information before generation from repair after failure.
- A mechanism-overlap audit that prevents symbolic planning and symbolic fallback being counted as independent gains.
- Task-paired, model-stratified inference with frozen boundaries, one outcome authority and explicit confirmatory/post-hoc claim labels.

Two-Stage Control Framework

STAGE 1 PLAN	**Goal + Environment State** objects · predicates · goals · action signatures	→	**Planning-Time Constraint** minimal → structured → engineered → Flat RAG → GraphRAG
	Candidate action sequence ↓ concrete validator signal		
STAGE 2 RECOVER	**Closed-Loop Response** open loop · reflection · frozen memory · PDDL recovery	→	**Official Evaluation** task success · goal completion · execution success · errors

Separation principle: the local verifier is an intervention component. Only the pinned official VirtualHome evaluator produces reported outcomes.

Factorial Treatments

Planning-time knowledge constraint

ID	Information added before first generation
B0	Minimal instruction and output schema
P0-S	Structured objects, state, goals and action signatures
P0-E	P0-S plus fixed precondition/order checklist
P1	P0-E plus task-conditioned training demonstration
P2	Global training graph: entity linking, GNN/PPR and 3-hop path reranking

Execution-time closed-loop control

ID	Response after detected failure
H0	Open loop; no post-generation intervention
H2-R	Return concrete validation failure; replan once
H2-M	Reflection plus one frozen development repair example
H2-P	Replace failed plan using symbolic PDDL search

Experimental Protocol

Matched design 5 planners × 4 harnesses × 3 models; 84 fixed tasks in every current cell.	**Models** DeepSeek-V4-Flash, GPT-5.5 and GLM-5-Turbo.
Primary outcome Official task success rate; failures remain in the denominator.	**Inference** Family-clustered 95% CIs and exact paired McNemar tests.

Frozen evaluator: EAI commit `531c62f8`. Temperature 0; 2,048-token cap. Current P2 uses a completed 24-cell same-cohort replacement replication.

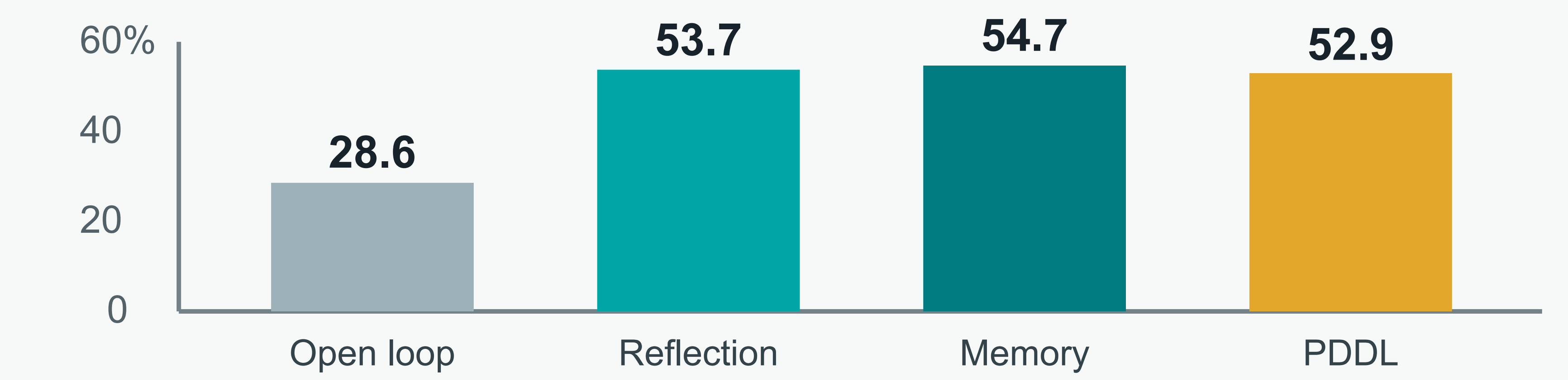
Threats to Validity

- Compatible subset, not the complete hidden EAI challenge set.
- Unseen task IDs but seen task families.
- VirtualHome Action Sequencing only; no BEHAVIOR result.
- Current GraphRAG evidence is post-hoc on the observed cohort, not untouched confirmation.

Experimental Results

76.2% best task success GPT-5.5 · B0 · reflection	**54.7%** highest harness mean frozen repair memory	**99.8%** PDDL mean execution success but only 52.9% task success

Recovery dominates task success (current 60-cell evidence package)



Task success is not the same as syntactic executability: symbolic recovery nearly eliminates execution errors but does not guarantee all task goals.

Planner × Recovery Interaction

Official task success (%): archived non-P2 + replacement P1/P2

	Open loop	Reflect	Memory	PDDL
B0 Minimal	0.0	61.1	64.7	52.4
P0-S Structured	22.6	53.2	52.4	53.6
P0-E Engineered	18.3	50.8	52.0	54.8
P1 Flat RAG*	49.6	50.8	50.8	50.8
P2 GraphRAG*	52.4	52.8	53.6	53.2

Diminishing returns: recovery adds up to **+76.19 percentage points** to an ungrounded baseline, but only 0–2.38 points on replacement Flat RAG. *P1/P2 rows are post-hoc same-cohort replication.

What the Results Mean

- **Recovery remains the strongest effect:** mean task success rises from 28.6% open loop to 53–55% with closed-loop control.
- **Current GraphRAG is promising, not confirmatory:** replacement P2 averages 53.0% vs 50.5% for rerun P1 on the observed cohort.
- **P2–P1 varies by model under open loop:** +4.76 pp (DeepSeek), +5.95 pp (GPT) and −2.38 pp (GLM); paired tests do not establish stable superiority.
- **Retrieval is template transfer:** 0/84 task-ID overlap, but 84/84 families are seen and 75/84 selected demos share the exact gold plan.

Conclusion & Next Steps

Where intervention occurs matters as much as which component is used.

Planning-time retrieval improves first-pass plans; execution-time recovery is most valuable when those plans are weak. Explicit evidence labels prevent a replacement implementation from being mistaken for untouched confirmation.

Next: confirm current GraphRAG on a new untouched cohort, evaluate the complete hidden EAI protocol, extend to BEHAVIOR and other modules, and add pricing for monetary cost reporting.

Selected References

[1] Xiang et al. *Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making.* arXiv:2410.07166, 2024.
[2] Puig et al. *VirtualHome: Simulating Household Activities via Programs.* CVPR, 2018.
[3] Shinn et al. *Reflexion: Language Agents with Verbal Reinforcement Learning.* NeurIPS, 2023.
[4] Lewis et al. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* NeurIPS, 2020.